



Evaluation of a spatially sex-specific assessment method incorporating a habitat preference model for blue marlin (*Makaira nigricans*) in the Pacific Ocean

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ABSTRACT

Blue marlin, widely distributed throughout the Pacific Ocean, are sexually dimorphic, have certain preferred habitats, and migrate seasonally. These characteristics have been ignored in previous stock assessment models. A population dynamics model that includes spatial structure, and sex and age structure was therefore constructed and fitted to fisheries data for blue marlin, along with information on the relative density of the population over space derived from a habitat preference model that uses the oceanographic and biological variables sea-surface temperature, mixed layer depth, sea-surface height anomaly, and chlorophyll-*a* concentration. Monte Carlo simulation was then used to examine the estimation performance of the stock assessment method. Estimates of management-related quantities including current spawning stock biomass are substantially biased when the assessment method ignores seasonal movement and sexual dimorphism. We also found that (i) uncertainty about the relationship between catch rate and abundance influences estimation performance to a larger extent than uncertainty in catches, (ii) the outcomes of the assessment are sensitive to the values assumed for natural mortality and stock-recruitment steepness, and (iii) the ratio of current spawning stock biomass to that at pre-exploitation equilibrium appears to be the most robust among the quantities considered. We

conclude that assessment methods for blue marlin in the Pacific Ocean need to take account of seasonal migration and sex structure to improve stock assessments.

Key words: evaluation of estimation performance, Monte Carlo simulation, oceanographic variables, seasonal movement, sexual dimorphism

INTRODUCTION

Blue marlin (*Makaira nigricans* Lacépède, 1802) is a cosmopolitan species distributed throughout tropical, sub-tropical, and temperate waters of the Pacific and Atlantic Oceans between 45°N and 45°S (Nakamura, 1985). It has been consolidated into a single species from two species in the Indo-Pacific and Atlantic Oceans based on analyses of genetic divergence (Maguire *et al.*, 2006). Molecular analyses (Graves and McDowell, 2003) and fishery catch rates suggest a single stock of blue marlin in the Pacific Ocean (Kleiber *et al.*, 2003). This assumption is also supported by the results of tagging experiments which have shown that blue marlin migrate throughout the Pacific Ocean (Richardson *et al.*, 2009). Blue marlin is a commercially important fish, and is primarily caught in pelagic longline fisheries targeting tunas, although small catches are also taken by gillnets and purse seine as well as by surface trolls, handlines and harpoons. It is the largest of the billfishes, attaining up to 450 cm in length and over 900 kg in weight, and is the most popular gamefish because of its size and fighting ability (Nakamura, 1985).

Blue marlin exhibit sexual dimorphism, with females being much larger than males (Wilson *et al.*, 1991). Several possible reasons have been proposed to explain this, including differential growth and mortality rates between the sexes (Hill *et al.*, 1989). The migration pattern of blue marlin has been inferred from seasonal catches in various fisheries (Hinton and Nakano, 1996; Walsh *et al.*, 2005). Seasonal movement could be related to shifts in sea surface temperature (SST) because blue marlin are found at the

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oceanic surface layer at temperatures above 26°C (Molony, 2005; Su *et al.*, 2008). Movements recorded by tags indicate that blue marlin migrate long distances, up to 5000 miles in some instances (Holland *et al.*, 1990; Richardson *et al.*, 2009). However, it is difficult to determine the movement paths and regular migration routes for individual fish throughout their life owing to low recapture rates. A habitat preference model for blue marlin developed by Su *et al.* (2008) using spatially and monthly resolved satellite-based oceanographic and biological data [i.e., SST, mixed layer depth (MLD), sea-surface height anomaly (SHA), and chlorophyll-*a* concentration (CHL)] indicates that there is seasonal variation in the distribution of blue marlin and that the migration pattern appears to be related to seasonal changes in habitat. The movement pattern of blue marlin is thus modeled using the habitat preference model.

Several stock assessments have been conducted for blue marlin in the Pacific Ocean. However, the status of this population is still uncertain because the results of the assessments range from the stock being close to fully exploited, to being in a healthy condition (Kleiber *et al.*, 2003). The abundance index for blue marlin developed by Ueyanagi *et al.* (1989) using recreational fisheries data shows an increasing trend from 1962 to 1985. However, an abundance index estimated using data from the Japanese longline fishery declined sharply since the early 1960s (Uosaki and Bayliff, 1999). Standardized catch-rate indices for blue marlin using deterministic habitat models (det-HBS) declined before 1970, but increased gradually thereafter (Hinton and Nakano, 1996). Therefore, it seems impossible to determine the status of this population based simply on a single fishery-based abundance index.

The exploitation status of blue marlin in the Pacific Ocean was assessed using a delay-difference model based on data from the Japanese longline fisheries for 1955–97 (Hinton, 2001). The population was assessed to be in a healthy condition, with current levels of biomass and fishing effort near those corresponding to maximum sustainable yield (MSY). Kleiber *et al.* (2003) assessed blue marlin in the Pacific Ocean using a statistical assessment method, MULTIFAN-CL, with the effective effort for the longline fisheries estimated using detHBS. This assessment concluded that the population was close to fully exploited because the biomass and exploitation rates were close to those needed to maintain MSY. However, those previous stock assessments of blue marlin ignored the possible impacts of seasonal movement and sexual dimorphism.

Simulations in which artificial data sets are generated and candidate assessment models are fitted to those data sets commonly have been used to evaluate the performance of fishery stock assessment methods (Punt *et al.*, 2002). Numerous simulation studies have evaluated the ability of production models and size- or age-structured methods of stock assessment (e.g., Punt, 2003; Wang *et al.*, 2005) to estimate quantities of interest to managers, such as MSY, the spawning stock biomass at which MSY is achieved (S_{MSY}), and the fishing intensity corresponding to MSY (F_{MSY}).

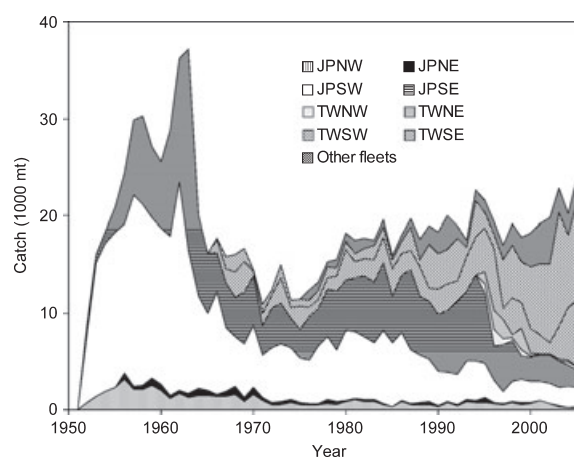
Here we use Monte Carlo simulation to explore the likely estimation performance of an assessment method that allows for seasonal movement and sexual dimorphism before this method is applied for actual stock assessments of blue marlin in the Pacific Ocean. Previous studies have not addressed assessment methods that primarily use length-frequency data and information from a habitat preference model, and thus the performance of this alternate assessment method cannot be inferred from the results of previous simulation evaluations of estimator performance. We apply this new assessment method in a variety of simulation trials to examine the extent to which violation of the assumptions of the method influence its performance.

MATERIALS AND METHODS

Data used

Many countries fish for blue marlin, but the bulk of the total catch is taken by Japan and Taiwan (Fig. 1). Therefore, the assessment model is based solely on the

Figure 1. Annual catches (1952–2006) of blue marlin from the Pacific Ocean by fleet (see Fig. 2 for the fleets).



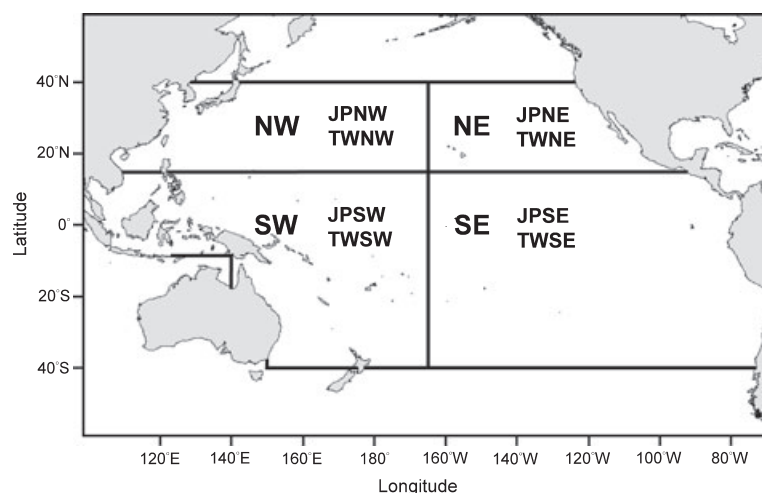


Figure 2. The Pacific Ocean showing the four regions considered in the analyses and the eight 'fleets' (four Japanese fleets – JPNW, JPNE, JPSW, and JPSE – and four Taiwanese fleets – TWNW, TWNE, TWSW, and TWSE).

data sets for the longline fleets of these two nations. The fishery data available for assessment purposes also relate to these pelagic longline fleets. Following Kleiber *et al.* (2003), the stock assessment of blue marlin in the Pacific Ocean is based on four regions divided at 15°N and 165°W (Fig. 2). The regions are defined primarily so that the Japanese coastal and offshore longline fisheries are then restricted to the northwest region. These four regions should be sufficient to adequately capture the spatial differences in density and seasonal movement patterns of blue marlin, yet not be too many that the number of estimable parameters becomes prohibitively large.

Catch and effort data are available for the Japanese and the Taiwanese longline fisheries (1952–2006 and 1966–2006, respectively). However, data are only available from 1956 for one of the Japanese fleets (JPNE), and from 1995 for two of the Taiwanese fleets (TWNW and TWNE; Fig. 2). The data for each catch-effort record includes information on time and location (5° × 5° blocks), the catch in number of fish, the effort in hooks, and the operating type. The nominal catch rates (defined as the number of fish caught per 1000 hooks) are standardized using a two-stage generalized linear modeling approach (i.e., the delta GLM method) to deal with the fact that some 5° × 5° blocks have no catch of blue marlin, but some effort (see Su, 2010 for details).

Length samples of blue marlin are treated differently from the catch and effort data, and are aggregated into 5-cm-long bins bimonthly. Length-frequency data for blue marlin are available for the Japanese longline fisheries from 1971 and for the Taiwanese longline fisheries from 1981. However, sex-specific length-frequency data are only available for the Japanese longline fisheries (the JPSE fleet) after 1991. No sex-specific length-frequency data are

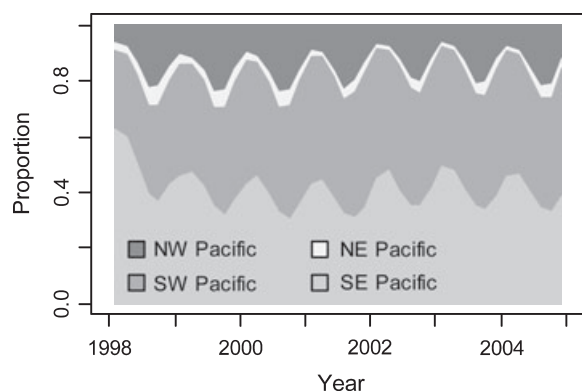
available for the Taiwanese longline fisheries. Therefore, length data without sex information for these two fisheries are aggregated over sex when used to estimate model parameters.

Movement can be estimated using capture-recapture data (e.g., Onitsuka *et al.*, 2010). However, the tagged animals only represent a small proportion of the population, which makes it difficult to determine movement patterns at the population level. Su *et al.* (2008) used generalized additive models (GAMs) to examine the relationships between spatio-temporal patterns of blue marlin abundance and related environmental factors (e.g., SST, MLD, SHA, and CHL) and to infer migration patterns from the estimated seasonal shifts in the spatial distribution. The seasonal movement of blue marlin is consistent over years, with slight annual variations, and appears to be related primarily to seasonal shifts in SST (Su *et al.*, 2008). The predicted relative densities of blue marlin by 5° × 5° blocks, summed bimonthly for each region, were used to compute proportions bimonthly and yearly (Fig. 3) which were in turn used as data when fitting the stock assessment model and hence estimating the parameters which determine movement.

Evaluating estimation performance using simulation

Following Punt *et al.* (2002), the steps in evaluating the performance of stock assessment methods using Monte Carlo simulation are: (i) develop a population dynamics model for the true population (i.e., the operating model; see Appendix A) and specify the values for some of its parameters based on the auxiliary information; (ii) fit the operating model for the situation in question to obtain values for the remaining parameters of this model (the results of this model fit provide the 'true' values for the quantities of management interest, listed below); (iii) generate artificial

Figure 3. The proportions of the blue marlin population in each of the four regions inferred from the habitat preference model of Su *et al.* (2008).



data sets (see Appendix B) by sampling from the predictions or adding noise to the predictions obtained by fitting the operating model to the data (step ii); (iv) fit the estimation model (taken here to be the same as the operating model) to the artificial data sets generated at step iii to provide estimates for the quantities of interest; (v) repeat steps iii and iv many times (100 simulations are sufficient to enable the effects of different factors to be distinguished); and (vi) compare the estimates and the true values using appropriate summary statistics.

The population dynamics model

Blue marlin are known to be sexually dimorphic. Therefore, the model of blue marlin population dynamics is sex-specific and age-structured (ages 0–15, with age 15 treated as a ‘plus group’), with different growth parameters and length-weight relationship for each sex. The model is based on a bimonthly time step and annual recruitments occurring at the start of the year. The age structure of the population at the start of 1937 is calculated under the assumption that the population was in equilibrium with respect to an estimated initial exploitation rate for each region at this time. In addition, blue marlin, as a highly migratory species, inhabit certain preferred habitats in the open ocean and migrate seasonally (Richardson *et al.*, 2009). Consequently, the population dynamics model is spatially structured. Movement of the population was modeled by the probability of moving from one region to another during each time step, with the parameters governing movement estimated primarily from the relative densities inferred from the habitat preference model of Su *et al.* (2008). The population dynamics model assumes that recruitment is related to spawning stock

biomass according to a Beverton–Holt stock-recruitment relationship (Beverton and Holt, 1957) and that the deviations about this relationship are log-normally distributed. Selectivity is assumed to be a logistic function of length.

Parameter estimation

The model is fitted to determine the values for the parameters of the operating model and also when testing the performance of the estimation model. The parameters of the population dynamics model can be divided into those for which auxiliary information is available (Table 1) and those which need to be estimated from the monitoring data (Table 2; Appendix C). The bimonthly data available for assessment purposes are: (i) the catches by region (assumed known without error, even though this may not be case), (ii) the catch-rate-based indices of abundance by region, (iii) the catch-length frequencies by fleet and sex (pooled across sex in some instances), and (iv) the proportion (relative density) of the population by region derived from the habitat preference model of Su *et al.* (2008).

Table 1. The values assumed for the parameters of the relationships between length and weight, maturity and age, and length and age, and for the standard deviation of the length-at-age.

Parameter	Female	Male
Length-at-50%-maturity, L_m (cm)	175.16	–
Maturity slope, r_m	–0.125	–
Length-weight, A	1.43×10^{-5}	1.12×10^{-5}
Length-weight, B	3.00	3.03
Asymptotic length, L_∞ (cm)	312.5	232.8
Growth parameter, k (per year)	0.111	0.131
Age-at-zero-length, t_0 (year)	–2.42	–3.58
Standard deviation of length-at-age, σ_a		
Age 1	7.75	4.80
Age 2	7.14	4.61
Age 3	6.63	4.45
Age 4	6.21	4.31
Age 5	5.85	4.20
Age 6	5.55	4.10
Age 7	5.29	4.01
Age 8	5.07	3.94
Age 9	4.89	3.88
Age 10+	4.72	3.88

Source: Sun *et al.* (2009), Dai (2002).

Table 2. The parameters of the population dynamics model not known from auxiliary information.

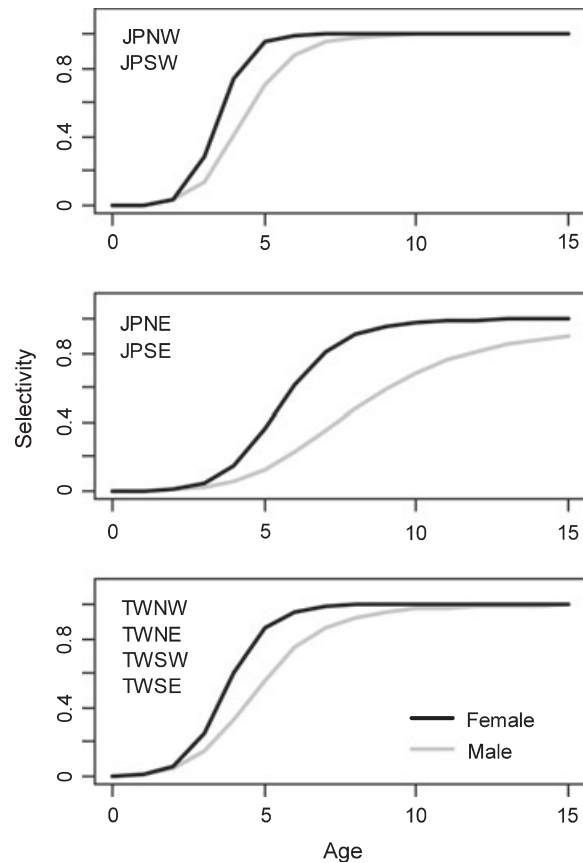
Parameter	No. of parameters
Estimated	
Unfished recruitment, R_0	1
Initial recruitment, R_{mit}	1 per year for 1937–51
Initial fishing mortality, F_{mit}	1 per region
Length-at-50%-selectivity, L_{50}	1 per fleet
Length-at-95%-selectivity, L_{95}	1 per fleet
Process errors, v_t	1 per year for 1952–2006
Moving matrix, X	12 per time-step
Prespecified	
Natural mortality, M (per year)	1
Steepness, h	1
Variation in recruitment, σ_v	1

The values for the parameters related to natural mortality (M), the steepness of the stock-recruitment relationship (h), and the extent of variation in recruitment (σ_v) cannot be determined from auxiliary information, nor can they be estimated reliably by fitting the model to the data, and they must therefore be prespecified (Magnusson and Hilborn, 2007). The values for M and σ_v are taken to be 0.3 per year (Kleiber *et al.*, 2003) and 0.4 (Wang *et al.*, 2005), respectively per year, respectively; h is assumed to be 0.5. This choice of h is a compromise between no compensation (i.e., recruitment is proportional to biomass) and no reduction in recruitment when biomass drops to a low level (i.e., recruitment is independent of biomass). In any case, values for h below 0.5 are not normally considered in stock assessments (Francis, 1992).

Initial results obtained when fitting the operating model to the data showed that the selectivity patterns for the four Taiwanese fleets were similar but that selectivity differed among fleets for the Japanese fleets. Estimating time-varying as well as dome-shaped selectivity led to only subtle differences in results; in particular, the fits to the length-frequency data were not markedly better even though many more parameters were estimated. Therefore, to achieve a parsimonious model, it was assumed that the length-specific selectivity pattern was logistic and time-independent, the same for all of the Taiwanese fleets, the same for the JPNW and JPSW fleets, and the same for the JPNE and JPSE fleets (Fig. 4).

Scenarios

Three sets of simulations (see Table 3 for a full summary) were conducted to examine the impact of dif-

Figure 4. Estimated age-specific selectivity ogives for the eight fleets.

ferent sources of uncertainty on estimation performance.

1 Simulations in which only one source of data (catch, catch rate, length frequency, and relative density) is subject to uncertainty. Four levels of observation error were examined for the catch data and the noise about the relationship between catch rate and abundance, and four effective sample sizes for length-frequencies and relative densities were examined. The values for the observation error standard deviations for the catch-rate data ($\hat{\sigma}_1^{A,f,m}$) were based on fitting the operating model to the data (see Eqn C.1). These values were then multiplied by 0.1, 0.5, 1, and 2 to examine different levels of uncertainty. Various plausible levels were chosen for the values assumed for the observation error standard deviations for the catch data ($\hat{\sigma}_C^{A,f,m} = 0.05, 0.1, 0.15, \text{ and } 0.2$) and for the effective sample sizes for the length-frequency ($n^L = 10, 20, 30, \text{ and } 40$) and the proportion data ($n^P = 10, 30, 50, \text{ and } 70$) likely to bound the actual values of these parameters.

Table 3. Simulations conducted to examine the impact of different sources of uncertainty on estimation performance.

	Parameters of the estimation model	Observation error
(1) Data subject to observation error		
Catch	Same as the operating model	$\hat{\sigma}_C^{A.f,m} = 0.05, 0.1, 0.15, \text{ and } 0.2$
Catch rate		Multiplier on $\hat{\sigma}_I^{A.f,m} = 0.1, 0.5, 1, \text{ and } 2$
Length frequency		$n^L = 10, 20, 30, \text{ and } 40$
Relative density		$n^P = 10, 30, 50, \text{ and } 70$
(2) Fixing parameters to wrong values		
M low (A)	M = 0.25 per year	$\hat{\sigma}_C^{A.f,m} = 0.05,$ Multiplier on $\hat{\sigma}_I^{A.f,m} = 0.1,$ $n^L = 40,$ and $n^P = 70$
M high (B)	M = 0.35 per year	
h low (C)	h = 0.3	
h high (D)	h = 0.7	
h higher (E)	h = 0.9	
σ_v low (F)	$\sigma_v = 0.2$	
σ_v high (G)	$\sigma_v = 0.6$	
(3) Impacts of ignoring movement and sexual dimorphism		
Base assessment (SM)	Same as the operating model	$\hat{\sigma}_C^{A.f,m} = 0.05,$ Multiplier on $\hat{\sigma}_I^{A.f,m} = 0.1,$ $n^L = 40,$ and $n^P = 70$
No movement (NM)	Diagonal movement matrix	
No sexual dimorphism (NS)	Sex-aggregated length-at-age and weight-at-age	
No sexual dimorphism or movement (NN)	Sex-aggregated length-at-age and weight-at-age and a diagonal movement matrix	

2 Simulations in which the values for some of the prespecified parameters (natural mortality, stock-recruitment steepness, and the extent of variation in recruitment) were set to values ($M = 0.25$ and 0.35 per year; $h = 0.3, 0.7, \text{ and } 0.9$; $\sigma_v = 0.2$ and 0.6) that differ from those assumed by the operating model ($M = 0.3$ per year; $h = 0.5$; $\sigma_v = 0.4$). The lowest extent of observation error for the four data sources ($\hat{\sigma}_C^{A.f,m} = 0.05$, multiplier on $\hat{\sigma}_I^{A.f,m} = 0.1$, $n^L = 40$, and $n^P = 70$) was used when generating the data sets so that the impact of assuming the wrong values for the prespecified parameters could be detected.

3 Simulations in which the estimation model ignored movement dynamics, sexual dimorphism, and both movement and sexual dimorphism. Assessments that ignored sexual dimorphism were based on sex-aggregated biological parameters, i.e., length-at-age and weight-at-age set to the average of those for males and females. Movement was ignored in the estimation model by setting the movement matrix to be diagonal and excluding the data on the proportions of the population by region from the objective function. The 'true' values for the quantities of management interest

for these simulation trials were taken to be those based on the operating model (which incorporates movement and sexual dimorphism). The extents of observation error for the catch, catch-rate, length-frequency, and proportion data were set to the lowest levels, as for the second set of simulations.

Summarizing estimation performance

Key quantities of fisheries management interest used to evaluate the ability of the estimation model are listed in Table 4. We use the relative error to quantify the estimation ability for each replicate for the nine quantities of management interest: and expressed it as:

$$E_i^j = \frac{\hat{Q}_i^j - Q^j}{Q^j} \times 100, \quad (1)$$

where E_i^j is the relative error (%) for simulation i and quantity j , $\hat{Q}_i^j$ is the value for quantity j provided by the estimation method for simulation i , and Q^j is the 'true' value for quantity j . The median relative error (MRE) and the inter-quartile range (IQR) are used to represent the distributions of relative errors for each quantity of interest and each scenario. These two

Table 4. Key quantities of fisheries management interest used to evaluate the performance of the estimation model.

Symbol	Unit	Description
S_0	Weight (m)	Spawning stock biomass at pre-exploitation equilibrium
S_{2006}	Weight (mt)	Spawning stock biomass at the end of the last year of the assessment period
F_{2006}	Year ⁻¹	Fleet-aggregated fishing intensity* during the last year of the assessment period
MSY	Weight (m)	Maximum sustainable yield†
S_{MSY}	Weight (m)	Spawning stock biomass at which MSY is achieved
F_{MSY}	Year ⁻¹	Fishing intensity corresponding to MSY
S_{2006}/S_0	–	Ratio of S_{2006} to S_0
S_{2006}/S_{MSY}	–	Ratio of S_{2006} to S_{MSY}
F_{2006}/F_{MSY}	–	Ratio of F_{2006} to F_{MSY}

*See Eqn A.15 for the definition of this quantity.

†See Shepherd and Pope (2002) for details.

statistics were chosen because they are robust to extreme estimation outcomes and because they can measure both accuracy and precision. For example, estimates can be considered to be unbiased and precise when the MRE and the IQR are both close to zero.

RESULTS

Impacts of the observation error

In general, observation error on catches has minor impacts on model performance. The estimates for all quantities are almost identical to the true values when the level of observation error for the catch data is

assumed to be 0.05 (MREs between -1.3 and 0.7% ; Fig. 5). As expected, the precision of the estimates of the nine quantities is poorer when the level of observation error is greater (IQRs between 0.6 and 5.2% for $\hat{\sigma}_C^{A,f,m} = 0.15$). The nine quantities can be determined accurately (MREs between -2.4 and 0.5%), although less precisely, when the catch data are subject to the highest level of observation errors considered (i.e., $\hat{\sigma}_C^{A,f,m} = 0.2$).

The impacts of the lowest level of observation error for the catch-rate data (multiplying each $\hat{\sigma}_I^{A,f,m}$ by 0.1) are inconsequential for all quantities (MREs between -1.6 and 1.5% ; Fig. 6). The accuracy and precision of

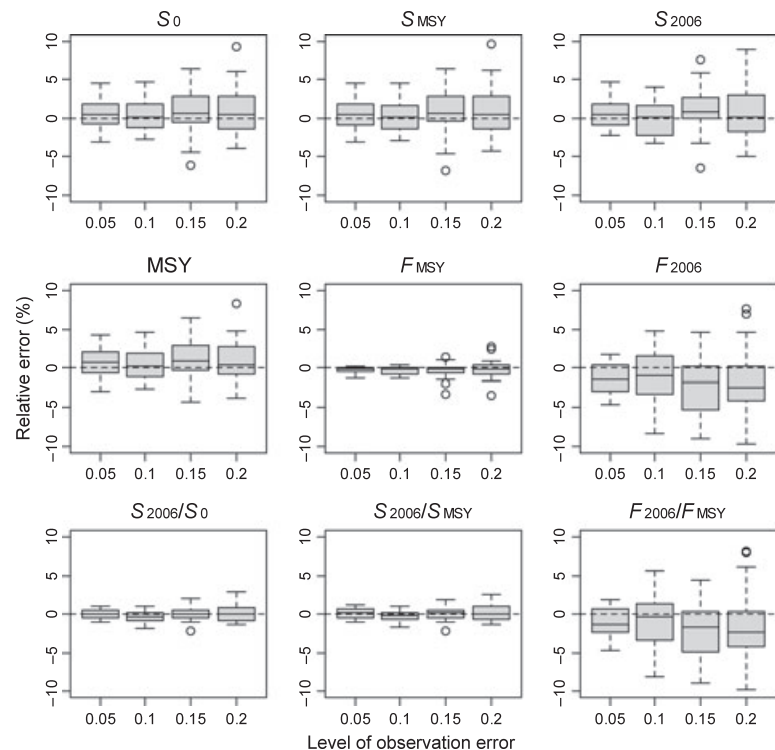


Figure 5. Box plots of the relative errors for the quantities of management interest corresponding to various levels of observation error for the catch data. The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

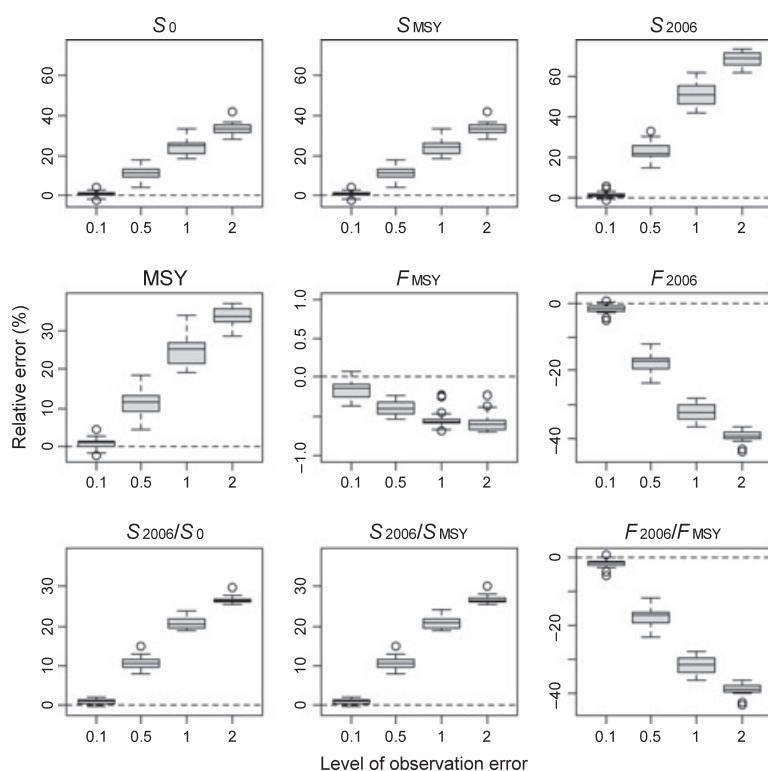


Figure 6. Box plots of the relative errors for the quantities of management interest corresponding to various levels of observation error associated with the catch-rate data. The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

the estimates and the fits to catch-rate data (results not shown) deteriorate linearly with increasing observation error for the catch-rate data. The original values of $\hat{\sigma}_{1f,m}^{Af,m}$ (i.e., a multiplier = 1) lead to greater level of bias and lower precision for all quantities (MREs between -32.1 and 50.7% and IQRs between 0.1 and 8.8%). The estimates of MSY and those related to biomass (S_0 , S_{MSY} , and S_{2006}) are positively biased when observation error is added to the catch-rate data. As a result, F_{2006} is negatively biased. However, F_{MSY} is estimated without much bias (MREs between -0.2 and -0.6%), probably because selectivity can be well determined. The two relative measures of spawning biomass (i.e., S_{2006}/S_0 and S_{2006}/S_{MSY}) are estimated somewhat better than the absolute values (S_0 , S_{MSY} , and particularly S_{2006}). However, F_{2006}/F_{MSY} is not estimated very well, because F_{MSY} is determined accurately but F_{2006} is not.

The estimates for each quantity of interest are biased and imprecise when the length-frequency sample size is small (MREs between -18.4 and 20.9% and IQRs between 9.3 and 37.6% for $n^L = 10$; Fig. 7). However, accuracy and precision improve substantially when the sample size is increased to 40 (MREs between -6.1 and 5.3% and IQRs between 3.1 and 12.9%). The estimates of the quantities related to population size (S_0 , S_{MSY} , and S_{2006}) are positively biased when the sample size is small (i.e., $n^L = 10$).

Similar patterns are evident for MSY , presumably because this quantity is associated with biomass. The estimates related to fishing intensity (F_{MSY} and F_{2006}) are also somewhat biased with a sample size of 10 (MREs between -11.3 and 6.8%). The estimates for F_{MSY} and F_{2006} are much more accurate and precise when the sample sizes are increased to 20 , 30 , and 40 (MREs between -3.9 and 3.2% and IQRs between 5.0 and 21.9%). As expected, the relative biomass measures are estimated more accurately than the absolute biomass measures, whereas the estimates of the ratio F_{2006}/F_{MSY} are negatively biased when $n^L = 10$; however, the extent of negative bias is less for larger sample sizes.

The estimates of the quantities related to biomass (S_0 , S_{MSY} , and S_{2006}) are negatively biased (MREs between -7.1 and -6.1%) when the sample sizes for the proportion data are small ($n^P = 10$; Fig. 8). However, the MREs for these quantities approach zero when the sample sizes for the proportion data are increased to 50 and 70 (MREs between -1.5 and -1.0% ; note the difference in scale between Figs 7 and 8). The pattern for MSY is similar to those for the quantities associated with biomass. The estimates for F_{MSY} are negatively biased when the sample sizes for the proportion data are small (MRE = -6.0% for $n^P = 10$), and become more accurate with larger sample sizes. The estimates for current fishing

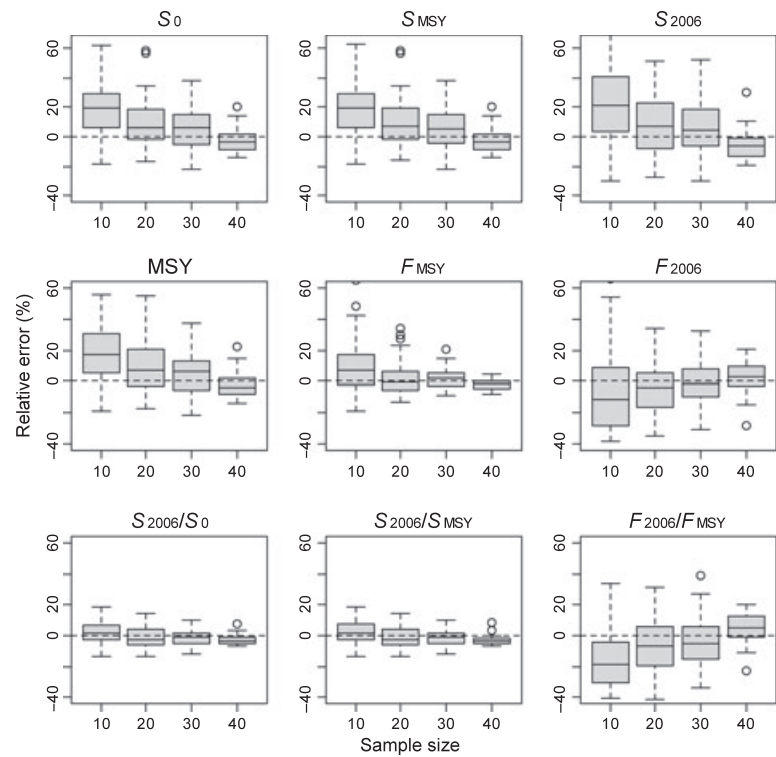


Figure 7. Box plots of the relative errors for the quantities of management interest corresponding to various sample sizes for the length-frequency data. The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

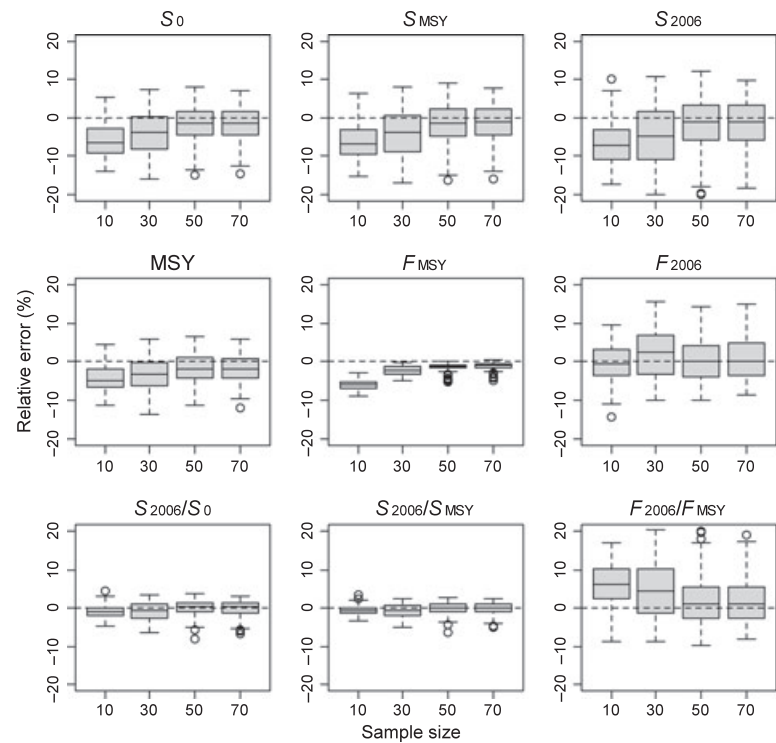


Figure 8. Box plots of the relative errors for the quantities of management interest corresponding to various sample sizes for the data on the proportion of the population in each area. The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

intensity, F_{2006} , are almost unbiased (MREs between -0.4 and 2.4%), although somewhat imprecise (IQRs between 6.7 and 9.8%). The two quantities related to

relative biomass are, as expected, estimated more accurately (MREs between -0.9 and 0.3%) than the measures of absolute biomass (MREs between -7.1

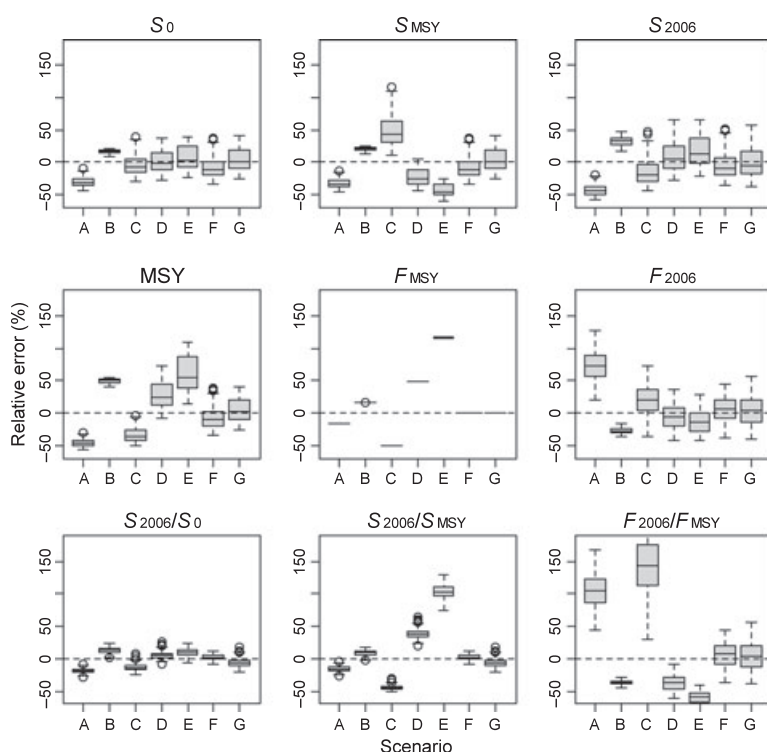


Figure 9. Box plots of the relative errors for the quantities of management interest for the analyses in which the assumed values for natural mortality, stock-recruitment steepness, and the extent of variation in recruitment differ from those in the operating model (scenario A: $M = 0.25$ per year; B: $M = 0.35$ per year; C: $h = 0.3$; D: $h = 0.7$; E: $h = 0.9$; F: $\sigma_v = 0.2$; G: $\sigma_v = 0.6$). The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

and -1.0%). However, the estimates of the ratio of current fishing intensity to F_{MSY} are positively biased (MRE = 6.1%) for a sample size of $n^P = 10$ because F_{MSY} is negatively biased.

Impacts of fixing parameters to wrong values

The sensitivity of the results is examined by setting the values for some of parameters (M , h , and σ_v) so that they differ from those in the operating model. The nine quantities of management interest are all sensitive to the value assumed for natural mortality, whereas only the MSY-related quantities (MSY, S_{MSY} , and F_{MSY}) are sensitive to the value assumed for stock-recruitment steepness (Fig. 9). In contrast, the results are relatively insensitive to the value assumed for the extent of recruitment variation.

Assuming a larger value for natural mortality than the true value (Fig. 9, case B) leads to positively biased estimates of S_0 and S_{2006} (MREs = 17.7 and 32.6% , respectively). MSY-related quantities (MSY, S_{MSY} , and F_{MSY}) are also positively biased when the assumed M is larger than the true value, whereas F_{2006} is negatively biased. The level of bias is substantially lower when estimates are expressed in a relative form (MREs = 12.9% for S_{2006}/S_0 and 9.0% for S_{2006}/S_{MSY}). However, F_{2006}/F_{MSY} is still poorly determined (MRE = -37.2%) because F_{2006} tends to be negatively biased, whereas F_{MSY} is positively biased. The opposite

pattern is evident if the assumed value for M is 0.25 per year rather than the true value of 0.3 per year (Fig. 9, case A).

The quantities related to absolute biomass (S_0 and S_{2006}) are generally well determined even if h is incorrectly specified (Fig. 9, cases C, D, and E). However, this is not the case for the MSY-related quantities (particularly MSY, S_{MSY} , and F_{MSY}). This is not surprising because h determines the relationship between recruitment and spawning stock biomass, and hence the perceived productivity of the population. In contrast to the situation for S_{2006}/S_0 , expressing biomass and fishing mortality relative to S_{MSY} and F_{MSY} , respectively, does not markedly improve accuracy (MREs between -59.8 and 143.6% for S_{2006}/S_{MSY} and F_{2006}/F_{MSY}) because S_{MSY} and F_{MSY} are strongly biased (MREs between -50.6 and 116.3%).

Impacts of ignoring movement and sexual dimorphism

The performance of the base stock assessment (which includes movement and sexual dimorphism) is compared with those of assessments which ignore movement, which ignore sexual dimorphism, and which ignore both movement and sexual dimorphism. As expected, estimation performance is best for the base assessment (case SM in Fig. 10). The between-simulation variation in estimates (but not bias) is lower when movement is ignored (cases NM and NN in Fig. 10).

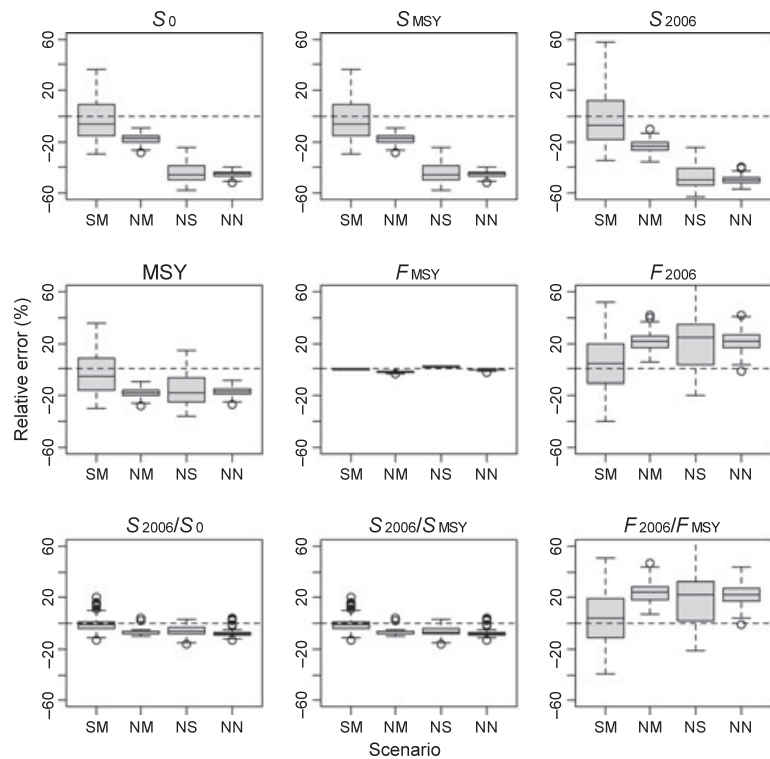


Figure 10. Box plots of the relative errors for the quantities of management interest when the estimation method ignores movement and sexual dimorphism (scenario SM: base stock assessment; NM: ignore movement; NS: ignore sexual dimorphism; NN: ignore movement or sexual dimorphism). The centerline represents the median and the box indicates the quartiles. The whiskers extend 1.5 times the inter-quartile range.

The estimates of S_0 , S_{MSY} , and S_{2006} are negatively biased when the estimation model ignores movement and when it uses sex-aggregated length-at-age and weight-at-age relationships. However, the larger of these two factors is sex (MREs between -45.1 and -49.9%). MSY is negatively biased and F_{2006} positively biased when movement and sex dimorphism are ignored. Estimates of the relative quantities are biased if either movement or sexual dimorphism is ignored. The estimates of relative biomass (S_{2006}/S_0 and S_{2006}/S_{MSY}) are again determined more accurately than those for absolute quantities (MREs between -7.9 and -6.3% for the cases NM, NS, and NN in Fig. 10). However, the accuracy of F_{2006}/F_{MSY} is not improved (MREs between 22.1 and 24.1%).

DISCUSSION

Impacts of the factors considered

Measurement error on the catch data seems inconsequential for the performance of the assessment model (Fig. 5). It could therefore be anticipated that the stock assessment method can estimate quantities of management interest reliably when the catch data are subject to random measurement error. However, discards occur in some fisheries due to the fish size or quotas, and, if available, should be included in the assessment (Punt

et al., 2006). Moreover, catches are often negatively biased (e.g., if discards are ignored) and this would be expected to lead to bias (underestimation of biomass) if the bias in reporting is constant over time, and to trends in bias if there is a trend in catch mis-reporting.

Uncertainty about the relationship between standardized catch-rate and abundance influences estimation performance to a larger extent than uncertainty in catches (Fig. 6). This is not surprising because the catch-rate data provide the most important source of information on the trend in abundance. Although the biases are negligible for a multiplier on $\hat{\sigma}_I^{A.f.m}$ of 0.1 , they can be much more substantial, e.g., MREs of 50% for S_{2006} , when the extent of observation error in catchability is increased to the level actually estimated for blue marlin. Although the estimation model is not mis-specified, increasing the precision of the catch-rate index leads to better fits to the catch-rate data (as expected) but also to poorer fits to the length-frequency and (particularly) the proportion data. In principle, the biases associated with the catch-rate data could be reduced substantially by reducing the noise associated with the catch-rate data for the Taiwanese fleets (e.g., Walsh *et al.*, 2005). This emphasizes the well known importance of obtaining an index of abundance that is precise and related to abundance in a linear (or known) way.

The results of the simulations suggest that the management of blue marlin in the Pacific Ocean should be based on relative biomass measures, e.g., S_{2006}/S_0 and F_{2006}/F_{MSY} , because these quantities can be estimated more accurately than absolute biomass measures even when the sample size for length-frequencies and the proportion data is small. For example, absolute biomass measures tend to be estimated with greater positive bias than relative biomass measures for an effective sample size for the length-frequency data of 10 (Fig. 7). Although the effective sample size cannot be related at present to the actual number of fish measured, increasing the effective sample size for length-frequency data to at least 20 for each combination of year, fleet, and sex can lead to nearly unbiased estimates for the quantities considered (Fig. 8). However, it should be noted that the effective sample size which leads to unbiased estimates is derived when there is no observation error except that related to relative density (also for length-frequencies).

Some of the parameters of the model can be determined based on biological research (e.g., length-at-age). However, M , h and σ_v are hard to estimate reliably using a population dynamics model because of the lack of informative abundance-index or age-composition data (see Magnusson and Hilborn, 2007), and thus need to be prespecified based on auxiliary information (Wang *et al.*, 2005). The simulations that explored the impact of error when prespecifying these parameters show that the results are sensitive to the values assumed for M and h , but are relatively insensitive to the value assumed for σ_v (Fig. 9).

In principle, estimates of natural mortality could be obtained from research such as tagging studies (e.g., Hampton, 2000). Although several tagging experiments have been conducted for blue marlin, those studies have focused on horizontal and vertical movement and post-release survival rates of the tagged animals (e.g., Holland *et al.*, 1990; Kerstetter *et al.*, 2003). Boggs (1989), Hinton (2001), and Dai (2002) used empirical equations to estimate natural mortality rates for blue marlin. However, their estimates for M range from 0.08 to 0.81 per year, whereas Kleiber *et al.* (2003) suggested that a value of 0.3 per year (the base value for M in this study) should be assumed when conducting stock assessments for blue marlin. Sex-specific natural mortality rates and regional growth rates should be considered in the stock assessment if reliable values for these parameters are available.

Steepness is used to specify the stock-recruitment relationship (Francis, 1992). High values for h (e.g., 0.9) have been assumed for pelagic apex predators

such as swordfish because expected recruitment appears to be relatively insensitive to changes in parental biomass (Wang *et al.*, 2005). However, it seems inappropriate to assume a high value of h for blue marlin in the Pacific Ocean because substantial drops in recruitment of this population have been observed at low adult abundance (Molony, 2005; Kitchell *et al.*, 2006). The value assumed for h in the base-case analysis (0.5) is a compromise between no compensation and no reduction in recruitment when biomass drops below 20% of the virgin biomass. The direction and quantitative magnitude of any bias associated with quantities of management interest will depend on the difference between the true and assumed values for h . However, the results of the simulations show that assuming an incorrect value for steepness does not have a great impact on the ability to estimate some management quantities (e.g., S_{2006}/S_0), but that quantities which are expressed relative to the biomass or exploitation rate corresponding to MSY can be very poorly determined (Fig. 9).

Spatially structured models that incorporate a movement matrix could take impacts of seasonal movement among regions into account (e.g., MULTIFAN-CL and the model of this study). However, Kleiber *et al.* (2003) noted that the movement parameters for blue marlin in the Pacific Ocean are poorly determined using MULTIFAN-CL because changes in catch-size compositions do not provide much information on movement. Incorporating movement into a population dynamics model involves determining the direction of the movement and the relative densities of the population among regions. Estimates of movement could be improved by including information from the results of tagging studies. However, the information from various conventional and archival tagging experiments is insufficient at present for determining the movement pattern of blue marlin (Kleiber *et al.*, 2003). In the absence of tagging data, the relative densities of the population among regions in different periods of the year, such as those derived from the habitat preference model with environmental factors relevant to blue marlin, were used to determine the seasonal movement pattern. The migration pattern of blue marlin in the Pacific Ocean is fairly consistent over the years for which data are available (Su *et al.*, 2008). However, increased variation in movement rates are likely during periods of enhanced environmental variability, such as El Niño Southern Oscillation (ENSO) events.

Billfish commonly exhibit sexual dimorphism (Boggs, 1989; Molony, 2005). This study confirms that estimates of biomass and fishing intensity are

substantially biased when sexual dimorphism is not taken into account, as suggested previously by Wang *et al.* (2005). The use of sex-aggregated biological parameters, i.e., those for length- and weight-at-age, in a population dynamics model ignores some major factors. For example, the biomass and the vulnerability to longline gears differ between the sexes for blue marlin (Fig. 4).

Implications for stock assessment and management

Most stock assessments are based on population dynamics models which assume that the stock being managed is homogeneous over the area fished and that differences in, for example, length frequencies for various fleets are due to differences in selectivity. Many stock assessments allow for sexual dimorphism, as is clearly warranted for blue marlin. However, very few stock assessment methods explicitly allow for the possibility of spatiality (MULTIFAN-CL and Stock Synthesis being noteworthy exceptions). Stock assessments which include a spatial component can be considered to be examples of 'extended stock assessments' (Plagányi, 2007), which are considered to be one way in which an ecosystem approach to fisheries (i.e., ecosystem based fisheries management) can be implemented (Hollowed *et al.*, 2011).

Extended stock assessments usually involve predation by species included in the assessment or environmental drivers of recruitment. In contrast, the assessment evaluated in this paper is 'extended' by allowing for spatial structure and integrating into the assessment a novel data source, the time series of the proportions of the population in various regions. The proportion data are inferred from a habitat model, and in turn are estimated using catch, effort, and relevant satellite-based oceanographic data at the scale of the ocean basin.

Oceanographic data are often used when standardizing catch and effort data to estimate indices of relative abundance for use in stock assessments (Maunder and Punt, 2004). This study highlights that these data can also be used to characterize drivers of spatial pattern of abundance which can then be included as a data source in stock assessments, augmenting the more traditional data sources such as catches, trends in standardized catch rates, and length compositions. This substantially reduces bias caused by ignoring spatial structure. The incorporation of other oceanographic variables that have been shown to be related to blue marlin density and distribution at finer scales, such as thermal fronts, current speed, and dissolved oxygen concentration (Prince and Goodyear,

2006) in the habitat preference model, may improve the predictive power of the model and hence the stock assessment.

An important aspect of fisheries stock assessment is understanding how environmental variability affects spatial dynamics of larvae and recruitment of a marine fish population. Several oceanographic variables, e.g., salinity, sea-surface temperature, and ocean current strength, have been demonstrated to influence population dynamics and recruitment variation directly by affecting survival of larvae or indirectly through planktonic prey species composition (Devine and Haedrich, 2011; Funamoto, 2011). A suite of oceanographic variables, e.g., sea-surface temperature and current velocity, were found to relate to the variation in larval tuna abundance, which then improves the estimates of current and projected biomass and exploitation rates (Langley *et al.*, 2009; Muhling *et al.*, 2010).

This study also illustrates the importance of precise indices of abundance (Fig. 6). In general, catch rates are standardized first and then used to fit the population model. Several approaches are available to develop standardized indices of abundance (see Maunder and Punt, 2004). Hinton and Nakano (1996) used deterministic habitat-based models to estimate relative indices of abundance for blue marlin. Unfortunately, to apply these models much more information is required about biology, habitats, and fisheries, such as the vertical distribution of the species, hydrographic data (e.g., temperature-at-depth), and the gear configuration (e.g., hook-at-depth), than is available for blue marlin.

Finally, the accuracy and precision of the nine quantities of management interest vary substantially depending on the specifications of the operating and estimation models. Relative biomass measures (S_{2006}/S_0 and S_{2006}/S_{MSY}) appear to be the least biased, although S_{2006}/S_{MSY} is markedly positively biased when h is overestimated, because S_{MSY} is negatively biased. That relative biomass measures are estimated more accurately confirms the conclusions from previous simulation studies based on biomass and age-structured models (e.g., Punt *et al.*, 2002; Punt, 2003; Wang *et al.*, 2005). This suggests that management for blue marlin in the Pacific Ocean should be based on the ratio of current biomass to S_0 or S_{MSY} because these two quantities can be estimated more reliably. In contrast, F_{2006}/F_{MSY} is poorly determined even though it is a ratio because F_{MSY} is estimated without much bias but F_{2006} is often substantially biased.

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APPENDIX A: POPULATION DYNAMICS MODEL

Basic population dynamics

The population dynamics of blue marlin are assumed to be governed by an age-, sex-, and region-structured model:

$$N_{t+1,a}^{s,A} = \begin{cases} N_{t+1,0}^{s,A} & \text{if } a = 0 \\ \tilde{N}_{t,a-1}^{s,A} & \text{if } 0 < a < \lambda, \\ \tilde{N}_{t,\lambda-1}^{s,A} + \tilde{N}_{t,\lambda}^{s,A} & \text{if } a = \lambda \end{cases} \quad (A1)$$

where $N_{t,a}^{s,A}$ is the number of fish of age a and sex s in region A at the start of year t , $\tilde{N}_{t,a}^{s,A}$ is the number of fish of age a and sex s in region A at the end of year t and λ is the maximum age considered (treated as a plus group).

The year is divided into six (bimonthly) time-steps to capture movement dynamics as well as natural and fishing mortality. Natural mortality is assumed to occur continuously throughout each bimonthly time-step, whereas fishery mortality is assumed to occur instantaneously in the middle of each time-step before movement. The number of fish of age a and sex s in region A at the start of bimonth $m + 1$ of year t , $\bar{N}_{t,m+1,a}^{s,A}$, is therefore calculated using the equation:

$$\begin{aligned} \bar{N}_{t,m+1,a}^{s,A} &= X_m^{A,A} \left(\bar{N}_{t,m,a}^{s,A} e^{-M\Delta t/2} - C_{t,m,a}^{s,A} \right) e^{-M\Delta t/2} \\ &+ \sum_{A' \neq A} X_m^{A,A'} \left(\bar{N}_{t,m,a}^{s,A'} e^{-M\Delta t/2} - C_{t,m,a}^{s,A'} \right) e^{-M\Delta t/2}, \\ \bar{N}_{t,1,a}^{s,A} &= N_{t,a}^{s,A} \text{ and } \tilde{N}_{t,a}^{s,A} = \bar{N}_{t,7,a}^{s,A} \end{aligned} \quad (A2)$$

where $X_m^{A,A'}$ is the probability that a fish in region A' at the end of bimonth m moves to region A at the start of bimonth $m + 1$ given that it survived bimonth m ($X_m^{A,A}$ is the probability that a fish stays in region A), M is the annual instantaneous rate of natural

mortality, Δt is the length of each bimonthly time-step (taken to 1/6 of a year), $C_{t,m,a}^{s,A}$ is the catch, in number, of fish of age a and sex s from region A during bimonth m of year t :

$$C_{t,m,a}^{s,A} = \sum_f C_{t,m,a}^{s,A,f} \quad (A3)$$

$C_{t,m,a}^{s,A,f}$ is the catch, in number, of fish of age a and sex s from region A by fleet f during bimonth m of year t .

Catches

The catch in number of fish of age a and sex s by fleet f in region A during bimonth m of year t is calculated using the equation:

$$C_{t,m,a}^{s,A,f} = F_{t,m,a}^{s,A,f} \left(\bar{N}_{t,m,a}^{s,A} e^{-M\Delta t/2} \right) \quad (A4)$$

where $F_{t,m,a}^{s,A,f}$ is the exploitation rate on fish of age a and sex s by fleet f in the region A during year t :

$$F_{t,m,a}^{s,A,f} = s_a^{s,f} \cdot F_{t,m}^{A,f} \quad (A5)$$

$s_a^{s,f}$ is the selectivity for fleet f on fish of age a and sex s , $F_{t,m}^{A,f}$ is the exploitation rate on fully selected animals by fleet f in region A during bimonth m of year t :

$$F_{t,m}^{A,f} = \frac{C_{t,m,obs}^{A,f}}{\sum_a \sum_s s_a^{s,f} \left(\bar{N}_{t,m,a}^{s,A} e^{-M\Delta t/2} \right)} \quad (A6)$$

$C_{t,m,obs}^{A,f}$ is the observed catch in number from region A by fleet f during bimonth m of year t .

Growth

The mean weight of an animal of age a and sex s is given by:

$$w_a^s = \sum_{l=1}^{\kappa} \Lambda_{a,l}^s \cdot A^s L_l^{B^s} \quad (A7)$$

where A^s , B^s , are the parameters of the length-weight relationship for a fish of sex s , L_l is the mid-point of length-class l , κ is the number of length-classes considered, $\Lambda_{a,l}^s$ is the fraction of animals of age a and sex s that are in length-class l :

$$\Lambda_{a,l}^s = \int_{L_l - \Delta L}^{L_l + \Delta L} \frac{1}{\sqrt{2\pi}\sigma_a^s} \exp \left[-\frac{(L - L_a^s)^2}{2(\sigma_a^s)^2} \right] dL \quad (A8)$$

ΔL is half the width of a length-class (2.5 cm), σ_a^s is the standard deviation of the length of a fish of age a and sex s , L_a^s is the expected length of a fish of age a and sex s :

$$L_a^s = L_{\infty}^s \left(1 - e^{-k^s(t-t_0^s)} \right) \quad (A9)$$

L_{∞}^s , k^s , and t_0^s are the parameters of the von Bertalanffy growth equation for fish of sex s .

Maturity

The proportion of female fish of age a which are mature is modeled using a logistic function:

$$\phi_a = \sum_{l=1}^{\kappa} \frac{\Lambda_{a,l}^{s=\text{female}}}{1 + \exp[r_m(L_l - L_m)]} \quad (\text{A10})$$

where ϕ_a is the fraction of females of age a that are mature, r_m is the maturity slope parameter, and L_m is the length-at-50%-maturity for females.

Recruitment

Recruitment, defined as number of fish of age 0, is assumed to be related to the spawning stock biomass by means of the Beverton–Holt stock-recruitment relationship, parameterized in terms of h , the steepness of the stock-recruitment relationship (Francis, 1992), and R_0 , the average recruitment in the absence of exploitation. The sex-ratio at age 0 is assumed to be 1 : 1 in the absence of evidence to the contrary. Density-dependence is assumed to be a function of the size of the entire mature biomass, i.e.,:

$$N_{t+1,0}^{s,A} = 0.5 \cdot x^A \cdot \frac{4hR_0S_t/S_0}{(1-h) + (5h-1)S_t/S_0} e^{v_t - \sigma_v^2/2} \quad (\text{A11})$$

where S_0 is the spawning stock biomass at unfished pre-exploitation equilibrium, S_t is the spawning stock biomass at the end of year t :

$$S_t = \sum_A S_t^A. \quad (\text{A12})$$

S_t^A is the spawning stock biomass in region A at the end of year t :

$$S_t^A = \sum_a \phi_a \cdot w_a^{s=\text{female}} \cdot \tilde{N}_{t,a}^{s=\text{female},A}. \quad (\text{A13})$$

x^A is S_{1937}^A/S_{1937} (1937 is selected because the population is assumed in unfished equilibrium at the start of 1937), v_t is a normally distributed process error, $v_t \sim N(0, \sigma_v^2)$, and σ_v^2 is the assumed variance of the process error in recruitment.

Selectivity

Selectivity-at-length is modeled using a logistic curve, and is assumed to be independent of sex. The assumption that the vulnerability of a fish increases monotonically to an asymptote with increasing length, has been made extensively extensively when representing selectivity for longline gears:

$$s_a^{s,f} = \sum_{l=1}^{\kappa} \Lambda_{a,l}^s \left(1 + \exp \left(-\ln 19 \frac{L_l - L_{50}^f}{L_{95}^f - L_{50}^f} \right) \right)^{-1} \quad (\text{A14})$$

where L_{50}^f is the length-at-50%-selectivity for fleet f , and L_{95}^f is the length-at-95%-selectivity for fleet f .

Initial conditions

It is impossible to project the model from pre-exploitation equilibrium because of a lack of catch records before 1952. Instead, following Hobday and Punt (2001), the population is assumed to have been in unfished equilibrium (projecting the model forward for 10 years, i.e., 1927–36) with respect to an estimated initial exploitation rate at the start of 1937, and the recruitments from 1937 are treated as estimable parameters. The age structure of the population at the start of 1952 is therefore determined by projecting the model from 1927 to 1951. The initial exploitation rate for each region is assumed to apply over the period 1937–51 for the purposes of this projection.

Fleet-aggregated fishing intensity

A measure of overall fishing intensity for the years in which all fleets took catches of blue marlin is calculated by dividing the total catch by a fleet-averaged exploitable number of animals in the middle of the year:

$$F_t = \frac{\sum_m \sum_A \sum_f C_{t,m,obs}^{A,f}}{\sum_A \sum_s \sum_a s_a^s (\bar{N}_{t,4,a}^{s,A})} \quad (\text{A15})$$

where F_t is the fleet-aggregated fishing intensity during year t , and s_a^s is the fleet-aggregated selectivity for fish of age a and sex s , calculated by weighting the fleet-specific selectivity patterns by the fleet-specific exploitation rates:

$$s_a^s = \frac{\sum_f \left(\sum_A \sum_t \sum_m F_{t,m}^{A,f} \right) s_a^{s,f}}{\sum_f \sum_A \sum_t \sum_m F_{t,m}^{A,f}} \quad (\text{A16})$$

where the summations over years relate to the years considered in the population dynamics model.

APPENDIX B: GENERATING ARTIFICIAL DATA SETS

The simulated data sets are created by adding observation error generated based on the distribution assumed when fitting each data source (except for the

catches which are assumed known when fitting the model) to the predictions of the operating model,¹ as follows.

1 The simulated catch data ($C_{t,m,obs}^{A,f}$) are generated by adding log-normally distributed observation error ($v_C^{A,f,m}; v_C^{A,f,m} \sim N(0, (\hat{\sigma}_C^{A,f,m})^2)$) to the 'true' catches:

$$C_{t,m,obs}^{A,f} = C_{t,m}^{A,f} e^{v_C^{A,f,m} - (\hat{\sigma}_C^{A,f,m})^2/2}. \quad (B1)$$

2 The simulated catch-rate data ($I_{t,m,obs}^{A,f}$) are generated by adding log-normally distributed observation error ($v_I^{A,f,m}; v_I^{A,f,m} \sim N(0, (\hat{\sigma}_I^{A,f,m})^2)$) to the 'true' catch-rates:

$$I_{t,m,obs}^{A,f} = I_{t,m}^{A,f} e^{v_I^{A,f,m} - (\hat{\sigma}_I^{A,f,m})^2/2}. \quad (B2)$$

3 The simulated length-frequency data for each combination of sex, region, year, and fleet ($P_{t,l,obs}^{s,A,f}$) are generated by sampling multinomially from the 'true' catch length-frequencies ($P_{t,l}^{s,A,f}$) with effective sample sizes (n^L) equal to $\rho^L n_t^{s,A,f}$ (see Eqn C.3), i.e., length-class l is selected with probability proportional to $P_{t,l}^{s,A,f}$.

4 The simulated proportion data ($p_{t,m,obs}^{A,f}$; the relative density of the population in each region) for each bimonth and year are generated by sampling multinomially from the 'true' proportion data ($p_{t,m}^A$) with effective sample sizes (n^P) implied by the value for ρ^P

recruitment relationship. The weighting factor (ρ^L) assigned to the length-frequency data in the likelihood function (see Eqn C.3) is set to 0.001. In contrast, the weighting factor for the proportion data (ρ^P) is set to 1000 to emphasize the contribution of proportion data in the likelihood function (see Eqn C.6). These weighting factors were selected by examining the model fits to the proportion and length-frequency data simultaneously. Estimating the weighting factors using the approach of McAllister and Ianelli (1997) leads to a value of 85 for the effective sample size for the length-frequency data, i.e., $\rho^L n_t^{s,A,f}$, which is larger than assumed when fitting the model. However, giving high weight to the length-frequency data leads to poor fits to the other data sources (the length-frequency sample sizes for some years are very large). Moreover, the approach of McAllister and Ianelli (1997) assumes that the residuals about the fits are independent, which is not the case here.

Catch-rate data

The indices of relative abundance based on standardized catch-rates are assumed to be log-normally distributed about the corresponding model predictions. The contribution of these data to the negative of the log-likelihood function, ignoring constants independent of the model parameters, is:

$$-\ln L_1 = \sum_f \sum_A \sum_t \sum_m \left\{ \ln \tilde{\sigma}_1^{A,f,m} + \frac{\left[\ln \left(I_{t,m,obs}^{A,f} \right) - \ln \left(q_{T(t),m}^{A,f} \cdot B_{t,m}^{A,f} \right) \right]^2}{2 \left(\tilde{\sigma}_1^{A,f,m} \right)^2} \right\} \quad (C1)$$

(see Eqn C.6), i.e., region A is selected with probability proportional to $p_{t,m}^A$.

APPENDIX C: CONTRIBUTIONS TO THE OBJECTIVE FUNCTION

The objective function minimized² to estimate the values for the parameters of the model includes two components: the contributions from data available for assessment purposes (i.e., catch-rates, length-frequencies, and relative densities across the region being assessed) and the constraints based on *a priori* assumptions for the deviations about the stock-

where $I_{t,m,obs}^{A,f}$ is the observed catch-rate (standardized) for fleet f in region A for bimonth m of year t , $q_{T(t),m}^{A,f}$ is the catchability coefficient for bimonth m of year t (where year t is in period T and $q_{T,m}^{A,f}$ might differ among periods; see below for details) for fleet f in region A , $B_{t,m}^{A,f}$ is the number of exploitable animals in the middle of bimonth m of year t in region A for fleet f after fishing:

$$B_{t,m}^{A,f} = \sum_s \sum_{a=1}^{\lambda} \left(s_a^{s,f} \cdot \bar{N}_{t,m,a}^{s,A} e^{-M\Delta t/2} - C_{t,m,a}^{s,A} \right) \quad (C2)$$

$\tilde{\sigma}_1^{A,f,m}$ is the standard deviation of the fluctuations in catchability for bimonth m and fleet f in region A , assumed to be year-invariant.

Maximum likelihood estimates exist for $\tilde{\sigma}_1^{A,f,m}$ and $q_{T,m}^{A,f}$.

It is often assumed that catchability is constant over time when conducting fisheries stock assessments

¹This data generation process only considers observation error and not process (recruitment) and model error. The impact of model error is explored by fitting variants of the estimation model which ignored sex and movement.

²Using AD MODEL BUILDER (<http://www.admb-project.org/>).

(Maunder and Punt, 2004). However, the catch-effort standardization does not account for all possible changes over time in fishing efficiency (Su, 2010). It is well known that catchability can vary over time because fishermen adjust their vessels, gear, and strategies in an attempt to maximize their catches (Hinton, 2001; Kleiber *et al.*, 2003), but none of these factors can be taken into account in the catch-effort standardization owing to lack of data. Therefore, the model application considers scenarios regarding the time period T over which catchability might be constant, taking account of information on when substantial changes in fishing practices occurred (such as the start of deep longlining operations for targeting bigeye tuna). Separate catchability coefficients are estimated for various periods of years for the Japanese longline fleets (1952–65 and 1966–2006), as well as the TWSW and TWSE fleets (1966–98 and 1999–2006) to account for changes over time in fishing practices, while catchability is assumed to be year-invariant for the TWNW and TWNE fleets (catch-rate data are only available for these fleets for 1995–2006).

Length-frequency data

Sex-specific length-frequency data (i.e., the fraction of fish by length class, summed to one over length and sex) are available and expressed as sex $s = 1$ for females and $s = 2$ for males. The sex-aggregated length-frequency data are denoted sex $s = 0$. The monthly length-frequency data are assumed to be multinomially distributed, and the contribution of the length-frequency data (ignoring constants independent of model parameters) to the negative of the logarithm of the likelihood function is (Punt *et al.*, 2006):

$$-\ln L_2 = \rho^L \sum_f \sum_A \sum_t \sum_m \sum_s n_{t,m}^{s,A,f} P_{t,m,l}^{s,A,f} \cdot \ln \left(P_{t,m,l}^{s,A,f} / p_{t,m,l}^{s,A,f} \right) \quad (C3)$$

where $P_{t,m,l}^{s,A,f}$ is the observed fraction that fish of sex s in length-class l made up of the catch (in numbers) by fleet f in region A during bimonth m of year t , $p_{t,m,l}^{s,A,f}$ is the model-prediction of the fraction that fish of sex s in length-class l made up of the catch (in numbers) by fleet f in region A during bimonth m of year t :

$$P_{t,m,l}^{s,A,f} = \frac{C_{t,m,l}^{s,A,f}}{\sum_{l'} C_{t,m,l'}^{s,A,f}} \quad (C4)$$

$C_{t,m,l}^{s,A,f}$ is the model-estimate of the catch (in numbers) of fish of sex s in length class l by fleet f in region A during bimonth m of year t :

$$C_{t,m,l}^{s,A,f} = \begin{cases} \sum_{a=1}^{\lambda} \Lambda_{a,l}^s \cdot C_{t,m,a}^{s,A,f} & \text{if } s = 1/2 \\ \sum_s \sum_{a=1}^{\lambda} \Lambda_{a,l}^s \cdot C_{t,m,a}^{s,A,f} & \text{otherwise} \end{cases}, \quad (C5)$$

ρ^L is the weighting factor assigned to the length-frequency data, and $n_{t,m}^{s,A,f}$ is the effective sample size for the length-frequency data for sex s , fleet f , region A , and year t .

Habitat indices of relative density

Given that standardized catch-rates could indicate relative abundance spatially, the relative densities of blue marlin abundance predicted from the habitat preferred model based on relevant oceanographic variables are included in the likelihood function under the assumption that the observed proportions of the population in each region are multinomially distributed about the corresponding model predictions. Ignoring constants independent of the model parameters, the contribution of the relative densities, to the negative of the logarithm of the likelihood function is:

$$-\ln L_3 = \rho^P \sum_t \sum_m \sum_A p_{t,m,obs}^A \cdot \ln \left(p_{t,m}^A / p_{t,m,obs}^A \right) \quad (C6)$$

where ρ^P is the weighting factor assigned to the data on the split of the population by region, $p_{t,m}^A$ is the model-estimate of the proportion of the population in region A at the start of bimonth m of year t (in number) based on the Japanese longline fleets (denoted by $f = 1$; the Japanese fleets were selected to

define the selected component of the population because they operated consistently over time and took the bulk of the catch of blue marlin in the Pacific Ocean):

$$p_{t,m}^A = \frac{\sum_s \sum_a s_a^{s,f=1} \cdot \bar{N}_{t,m,a}^{s,A}}{\sum_A \sum_s \sum_a s_a^{s,f=1} \cdot \bar{N}_{t,m,a}^{s,A}} \quad (C7)$$

$p_{t,m,obs}^A$ is the observed proportion of the population in region A at the start of bimonth m of year t (Fig. 3).

Constraint contribution to the objective function

A constraint is placed on the deviations about the stock-recruitment relationship for the entire period

considered in the population dynamics model (1937–2006):

$$\frac{1}{2\sigma_v^2} \sum_t v_t^2 \quad (\text{C8})$$

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